Title : DBT + Snowflake ETL

Fundamentals & Setup and Web Application Deployment

1.Objective and Purpose of the System :

The objective of this system is to design and implement a modern, scalable, and automated ETL pipeline using dbt (Data Build Tool) and Snowflake within the domain of data engineering and big data. The system focuses on transforming raw, unstructured, or semi-structured data into structured, high-quality datasets that can be used for analytics and business intelligence.

The purpose of this system is to:

- Automate data transformation using dbt
- Improve data quality through testing and validation
- Enable scalable data storage and processing using Snowflake
- Reduce manual intervention in ETL workflows
- Provide a web-based interface for monitoring and visualization
- Support real-time or near real-time data analytics

2. Tools, Technologies, and Platforms Used:

Category	Tool/Technology	Purpose
Data Transformation	dbt (Data Build Tool)	Transform and model data using SQL
Data Warehouse	Snowflake	Store and query large-scale data
Programming Language	SQL	Write transformation queries
	Python	Integration and automation
Processing Engine	Snowflake Engine	Query execution and processing
Cloud Platform	AWS / Azure / GCP	Hosting and scalability
Web Framework	Streamlit / Flask	Build web dashboard
Orchestration (Optional)	Apache Airflow	Schedule and automate workflows
Version Control	Git / GitHub	Code management and collaboration

3. System Setup and Components:

The system is composed of multiple layers and components:

a) Data Sources

- Input data from APIs, CSV files, databases, or streaming sources

b) Snowflake (Database Layer)

- Acts as the central data warehouse
- Stores raw (staging), processed, and final data

c) dbt Environment

- Contains models, tests, and configurations
- Performs transformations using SQL

d) Servers and Services

- Snowflake virtual warehouse for computation
- Cloud storage (optional) for raw data
- Web server for hosting dashboard

e) Configurations

- dbt project setup (models, seeds, tests)
- Environment variables and credentials
- Role-based access control in Snowflake

4. System Design and Flow:

The system follows a layered architecture:

Step-by-Step Flow:

1. **Data Ingestion**
   Raw data is collected and loaded into Snowflake staging tables
2. **Data Transformation (dbt)**
   - Cleaning and filtering data
   - Joining datasets
   - Applying business logic
3. **Data Modeling**
   - Creation of fact and dimension tables
   - Structuring data for analytics
4. **Data Storage**
   Final processed data is stored in Snowflake
5. **Web Application Layer**
   Data is visualized using a dashboard (Streamlit/Flask)

System Modules:

- Data Ingestion Module
- dbt Transformation Module
- Data Warehouse Module
- Web Application Module
- Monitoring & Logging Module

Architecture Diagram

The Architecture Behind the Magic

At its core, dbt Projects on Snowflake introduces a new type of database object called DBT PROJECT. Think of this as a container that holds all your transformation logic, tests, and documentation in a versioned, manageable format. These objects live at the schema level, making them easy to organize and control access to through Snowflake's robust security model.

The system works by storing your dbt project files directly within Snowflake's internal staging area, organized by version. This means every change you make is tracked and can be rolled back if needed. The versioning system uses identifiers like VERSION$1, VERSION$2, and so on, with special aliases like LAST and FIRST for easy reference.

5. Performance, Scalability, and Reliability Considerations:

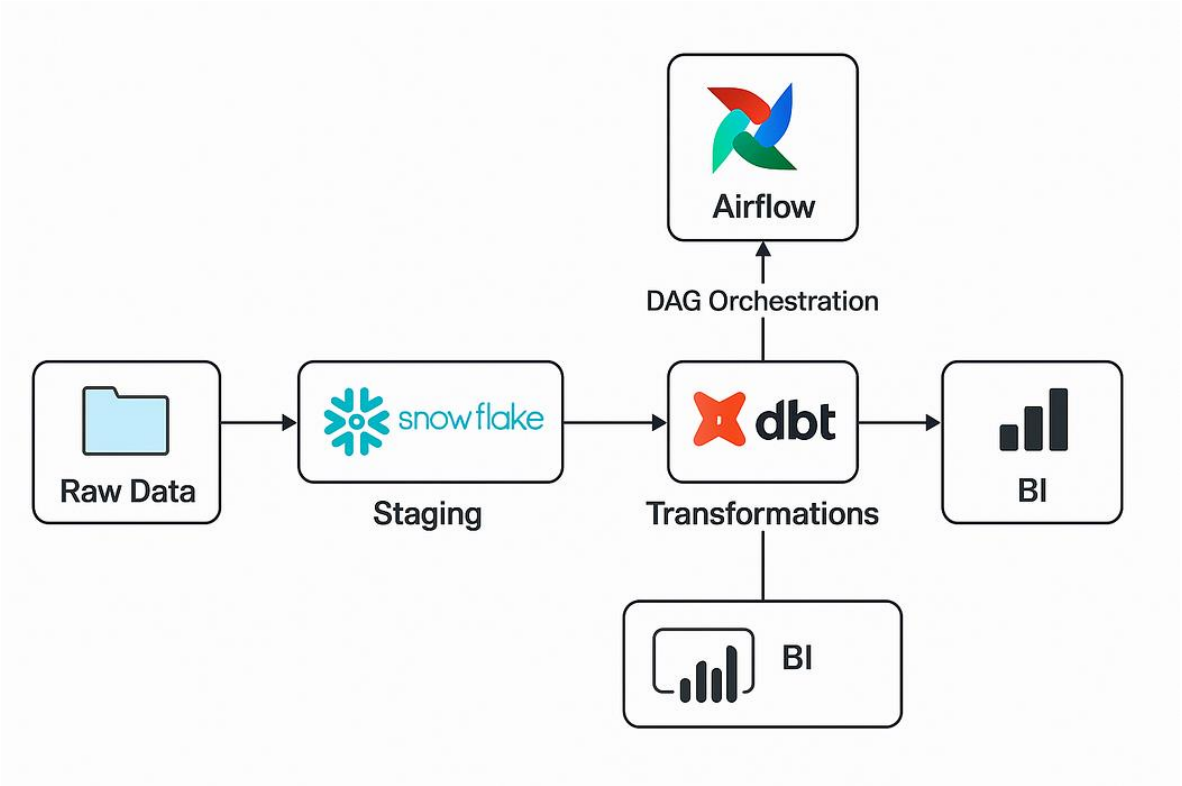
- **Performance:**
  - Use of distributed processing (Apache Spark)
  - Optimized queries in Snowflake
  - Incremental data processing using DLT
- **Scalability:**
  - Cloud-based infrastructure supports horizontal scaling
  - Snowflake auto-scaling warehouses
  - Ability to handle large datasets efficiently
- **Reliability:**
  - Automated error handling in pipelines
  - Data quality checks using DLT expectations
  - Backup and recovery mechanisms
  - Monitoring and alert system

6. Expected Outcomes, Diagrams, or Screenshots :

Expected Outcomes:

- Fully automated ETL pipeline
- High-quality, consistent, and reliable data
- Real-time data availability for analytics
- Interactive web dashboard for visualization

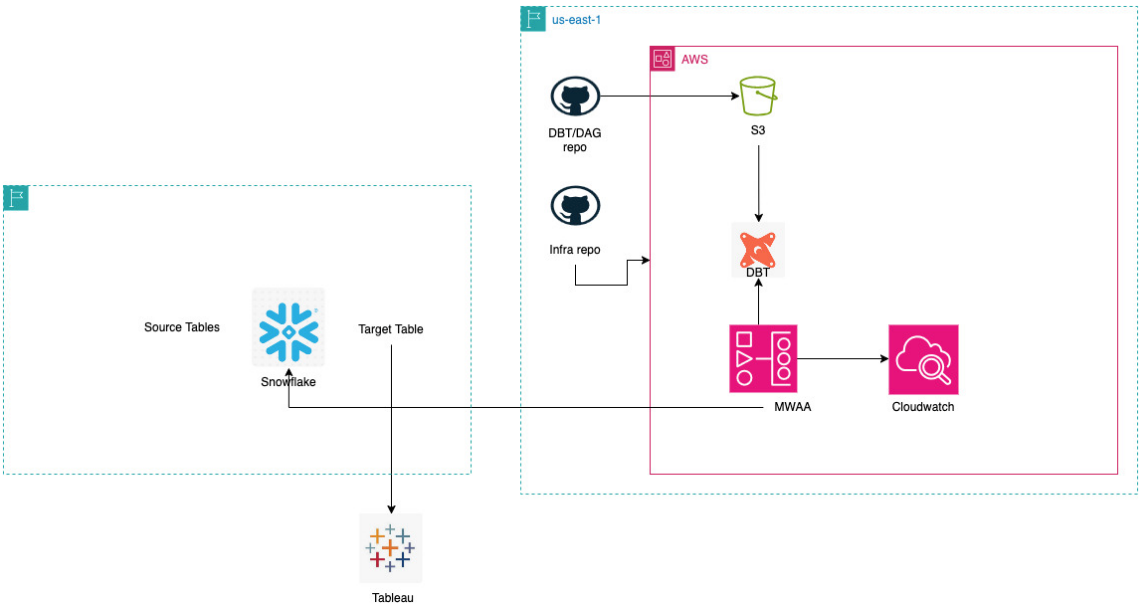
Building a Modern ELT Pipeline with dbt, Snowflake & Airflow



The Flow

```
Raw Data → Staging (Snowflake) → dbt Transformations → Fact Tables →
BI/Analytics
                ↑                                       │
                └──────────── Airflow DAG Orchestration ────────────┘
```

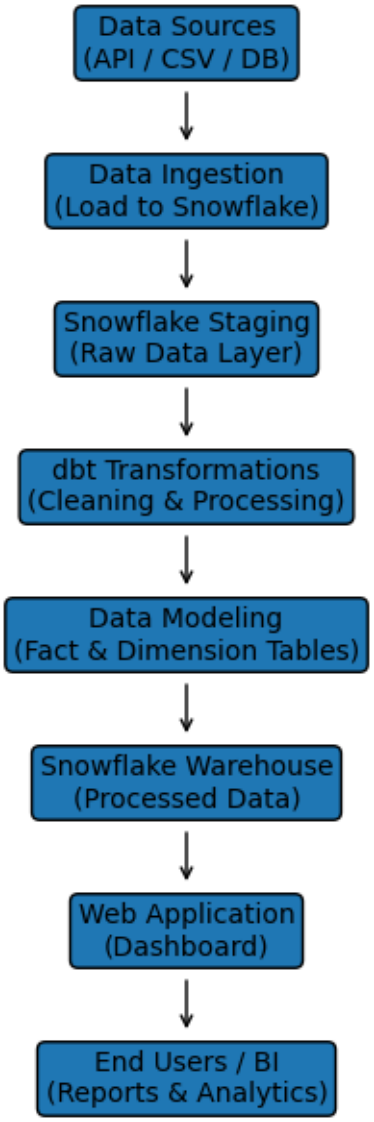
Modern ETL Architecture Diagram (for Report) :

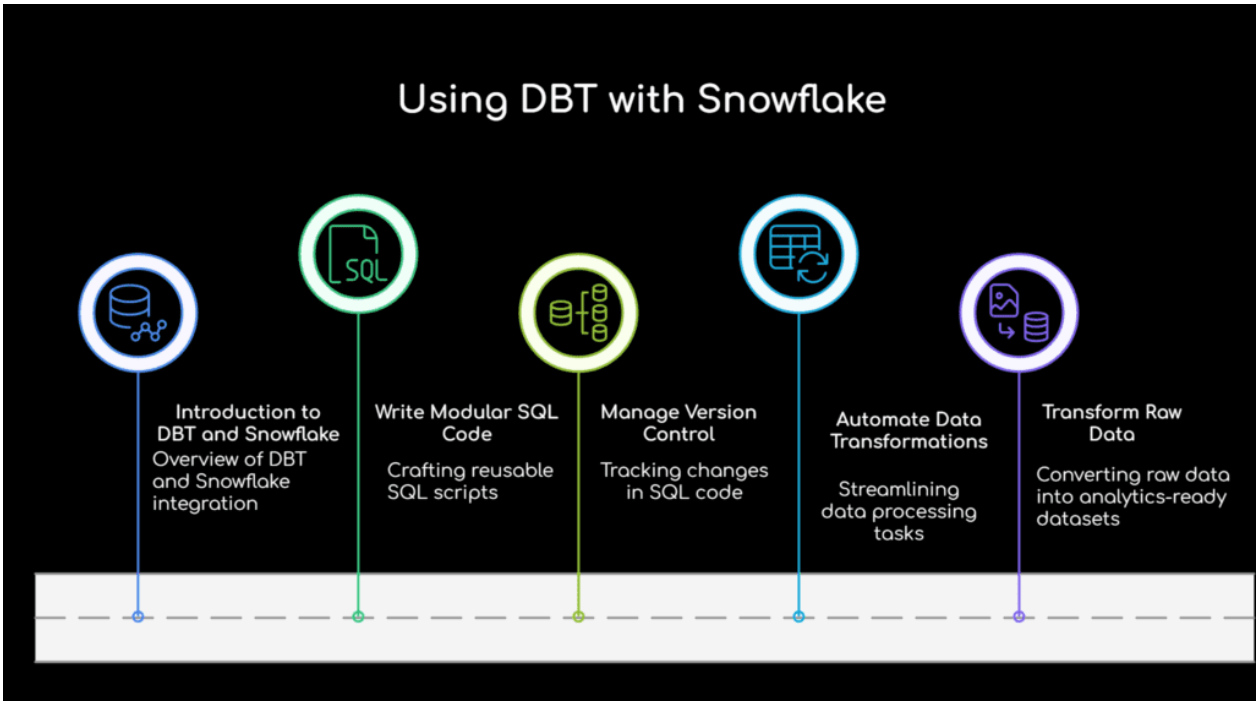


Once the S3 sync is completed, at schedule time, the DAG will be invoked and run dbt code. dbt commands such as `dbt run`, `dbt run –tag: [tag_name]`, etc., can be used.

With the run of dbt code, dbt will read the data from source tables in Snowflake source schemas and, after transformation, write to the target table in Snowflake. Once the target table is populated, Tableau reports can be generated on top of the aggregated target table data.

- The dbt repository will have two primary branches: main and dev. These branches are always kept in sync.
- Developers will create a feature branch from dev for their work.
    - The feature branch must always be rebased with dev to ensure it is up-to-date.
- Once development is completed on the feature branch:
    - A pull request (PR) will be raised to merge the feature branch into dev.
    - This PR will require approval from the Data Engineering (DE) team.
- After the changes are merged into the dev branch:
    - Github hooks will automatically sync the AWS-dev/stg AWS account's S3 bucket with the latest changes from the Git repository.
    - Developers can then run and test jobs in the dev environment.


Data Sources
(API / CSV / DB)
Data Ingestion
(Load to Snowflake)
Snowflake Staging
(Raw Data Layer)
dbt Transformations
(Cleaning & Processing)
Data Modeling
(Fact & Dimension Tables)
Snowflake Warehouse
(Processed Data)
Web Application
(Dashboard)
End Users / BI
(Reports & Analytics)


Using DBT with Snowflake
SQL
Introduction to DBT and Snowflake
Overview of DBT and Snowflake integration
Write Modular SQL Code
Crafting reusable SQL scripts
Manage Version Control
Tracking changes in SQL code
Automate Data Transformations
Streamlining data processing tasks
Transform Raw Data
Converting raw data into analytics-ready datasets